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LRFD methodology for river embankments against non-stationary flooding under climate change

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ABSTRACT

Riverine floods have become increasingly prevalent on a global scale, posing significant risks to infrastructure systems and communities. The escalating impacts of climate change associated with the increase in rainfall intensities and frequencies necessitate the improvement of the existing design methodologies to account for the non-stationary climate change effects to ensure that the reliability is above the target level and mitigate future flood disasters. This paper presents a novel LRFD approach for river embankments subjected to extreme rainfall under non-stationary climate change effects. This approach introduces an additional partial factor to account for the effects of climate change. Precipitation and temperature projections are collected from various climate models considering several cases of emission scenarios. An integrated hydrological and hydraulic modeling of the analyzed river is carried out to estimate the associated time-variant river discharge and water surface elevation. The non-stationary extreme value associated with the maximum flood level is leveraged using the peak-over-threshold approach. The embankment reliability and the corresponding most probable points are evaluated using limit states associated with overtopping and slope failures. Based on the estimated and target reliability indexes, the design point for each random variable is assessed considering the cases with and without climate change effects. Finally, the partial factors associated with climate change effects are determined. As an illustrative example, the proposed framework is applied to the Ashida River in Fukuyama city of Japan.

1. Introduction

In recent years, floods have increasingly become a global concern, posing a significant threat to infrastructure and communities worldwide [1–3]. Notably, Japan has been one of the regions most susceptible to flood-related risks, with a substantial portion of these events attributed to embankment failures. For instance, in September 2000, the heavy rainfall event in the Tokai district of Aichi Prefecture caused an overtopping of the embankment and affected 63,440 buildings, with 5.5 million people evacuated [4]. Similar catastrophic flood disasters stemming from embankment failures were also recorded in Niigata Prefecture in July 2004 [5], Kanto region in 2015 [6], and northern Kyushu in July 2017 [7]. More recently, in 2018, the western part of Japan was afflicted by an exceptionally high rainfall amount that caused embankment failures along the river near the residential area. This event

caused damage to over 2,000 buildings, resulted in 212 casualties, and triggered 619 geological disasters across 31 prefectures, marking it as one of Japan's most significant flood events since 1982 [8,9]. In addition to Japan, embankment failures that resulted in catastrophic flood events have been reported in the Netherlands [10], Indonesia [11], and Bangladesh [12]. These historical events indicate that riverine floods have been becoming more severe and are expected to worsen in the future due to climate change. Therefore, it is essential to design embankments with adequate reliability that can withstand the increasing intensity and frequency of future riverine flood events.

To mitigate riverine flood disasters, rainfall assessment is paramount since it forms the basis for evaluating hydraulic loads and overall stability associated with embankments. Extensive efforts have been made to develop a probabilistic rainfall model to assess the safety of embankments and geological hazards [13–19]. In response to the effects of

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climate change, various studies have introduced non-stationary extreme rainfall analyses [20–23]. Based on the probabilistic rainfall assessment, the hydrological and hydraulic integrated modeling approach was commonly adopted for embankment safety assessment using widely available tools such as the Soil and Water Assessment Tool (SWAT) and the Hydrologic Engineering Center River Analysis System (HEC-RAS) [24–27]. In addition, several studies have projected future riverine flooding scenarios and their potential intensification due to climate change using the aforementioned integrated approach [28–31]. However, these studies have not yet been implemented to develop appropriate embankment design methodologies considering climate change.

In its early stages of development, geotechnical design methods were primarily based on allowable stress design (ASD) [32–35]. In ASD, all uncertainties are taken care of by applying a safety factor on the resistance side, and the stresses are obtained using linear elastic analysis. ASD may occasionally lead to a design without sufficient reliability. The transition from ASD to probabilistic assessment of embankment became an uprising topic in the early 1970s after introducing the first-order reliability method (FORM) [36–40]. The semi-probabilistic approach of embankment design through load and resistance factor design (LRFD) leads to more efficient use of materials. Recently, the LRFD methodology has been developed and adopted for embankment by regulatory agencies such as the Missouri Department of Transportation [41,42].

Despite many efforts made to quantify the time-variant reliability of structures, the non-stationary effects of climate change associated with increasing intensity and frequency of extreme hazards have not yet been considered in the design methodology of embankment and slope structures [43–45]. Probabilistic embankment design can be carried out based on Monte Carlo simulation (MCS) [46,47]. Nonetheless, the MCS approach involves intricate computations, consumes an excessive amount of time, and is significantly impractical. Furthermore, the diverse temperature and rainfall projections from climate models and emission scenarios contribute to significant uncertainties in assessing future climate change impacts, in which many of these climate models exhibit discrepancies between the simulated climate conditions and the observed local climate characteristics [48,49]. Considering these challenges, a more practical design method is necessary through the development of LRFD methodology on structures under climate change effects to enhance future disaster preparedness.

This paper presents a novel framework for the LRFD methodology of river embankments by introducing an additional partial factor to accommodate the non-stationary impacts of climate change on embankment performance. Future precipitation and temperature projections are collected from various climate models considering several emission scenarios. To enhance the accuracy of climate data derived from global-scale climate models, Quantile Delta Mapping (QDM) is employed as a bias correction technique, allowing the refinement of the information for local conditions. An integrated hydrological and hydraulic analysis of the target river is then conducted to estimate the river discharge and water surface elevation over time. Based on the time-varying water surface elevation data, a non-stationary extreme value theory associated with the maximum water surface exceeding the river cross-section height (hereafter referred to as the maximum flood depth) is applied through the peak-over-threshold (POT) approach. The reliabilities of the embankment associated with the overtopping and slope failure modes are assessed. To illustrate the effectiveness of the proposed framework, an embankment in the Ashida River in Fukuyama city, Japan, is used as a study case. The impact of climate change on the embankment reliability and the corresponding partial factors are discussed.

2. Procedure for estimating partial factors for embankment design considering non-stationary climate change effects

Various studies have suggested that the changing climate would lead to more frequent and intense riverine flooding [50–52]. Due to the

effects of climate change, streamflow and the associated maximum water level are expected to increase with time. As a result, the increase in maximum water level would make the embankment more susceptible to failure due to overtopping and slope instability. Therefore, it is necessary to prevent future flood disasters by considering non-stationary climate change effects in embankment design methods to prevent future flood disasters.

Fig. 1 shows the procedure to determine the partial factors for embankment design considering non-stationary climate change effects. The method comprises five main steps: 1) data projection of future rainfall intensity and temperature from several climate models, 2) hydrologic and watershed modeling of the analyzed river, 3a) hydraulic modeling of river flow in the analyzed river, 3b) probabilistic modelling of water level excess, 4) reliability assessment of embankment, and 5) estimation of the partial factors for embankment design based on LRFD concept considering climate change. The details are described as follows.

In the first step, climate models, specifically General Circulation Models (GCMs) from the Coupled Model Intercomparison Project Phase 6 (CMIP6) [53], are collected under various climate change scenarios known as Shared Socioeconomic Pathways (SSPs) [54]. GCMs simulate large-scale global climate patterns but have limitations due to coarse spatial resolution (typically 100 to 300 km between grid points), hindering the capture of local-scale variations [55]. To address this, bias correction and statistical downscaling methods are applied. Detailed discussions of bias correction and statistical downscaling techniques are presented in Section 3.

Based on the bias-corrected temperature and precipitation data, the hydrological assessment of the river section is carried out in step (2). SWAT and ArcGIS are employed, utilizing geospatial data such as a digital elevation model (DEM), land use information, and soil type data to define the watershed boundary and assess Hydrological Response Units (HRUs) [56]. Step (3a) involves a one-dimensional hydraulic analysis using HEC-RAS simulation software from the United States Army Corps of Engineers [57]. The river geometry in the embankment is defined based on DEM data, and the unsteady flow analysis is performed using the river discharge output from step (2). In this study, the flood depth H_{exc} is defined as the difference between the water surface elevation above the river cross-section depth h_r and the ground level (see Fig. 2) [58]. Therefore, the maximum flood depth $H_{exc,max}$ should be evaluated to ensure the embankment can safely contain floodwaters and prevent overtopping events. Given the projected increase in flood intensity and frequency due to climate change, step (3b) introduces a method to combine POT using the non-stationary generalized Pareto distribution and extreme value analysis to estimate the time-variant distribution of the maximum flood depth from the present time t_0 to an analyzed time t_D (i.e., $F_{Hexc,max}(h_{exc}, [t_0, t_D])$). The details of steps (2) and (3) are described in Section 3.

In step (4), the reliability assessment associated with the embankment design parameters is evaluated considering two failure modes: overtopping and slope instability. In this study, the reliability of the embankment is evaluated from time t_0 to time t_D since the annual reliability approach for reliability-based design is not suitable for non-stationary load actions (further details are given in the [supplementary file](#)). The reliability of the embankment is significantly influenced by its geometry, particularly the phreatic line associated with the river's water level and the critical failure surface of the embankment. As a result, the limit state function of slope instability becomes increasingly complex. Therefore, in step (5), the partial factors associated with the embankment design variables are evaluated using the expanding ellipsoid approach introduced by Low and Tang [59], which is considered more beneficial in addressing the increasingly complex limit state function. Based on the evaluated reliability index, the design point for each embankment parameter is estimated in the proposed framework. The partial factors to ensure the safety of embankment against overtopping and slope instability can be defined as:

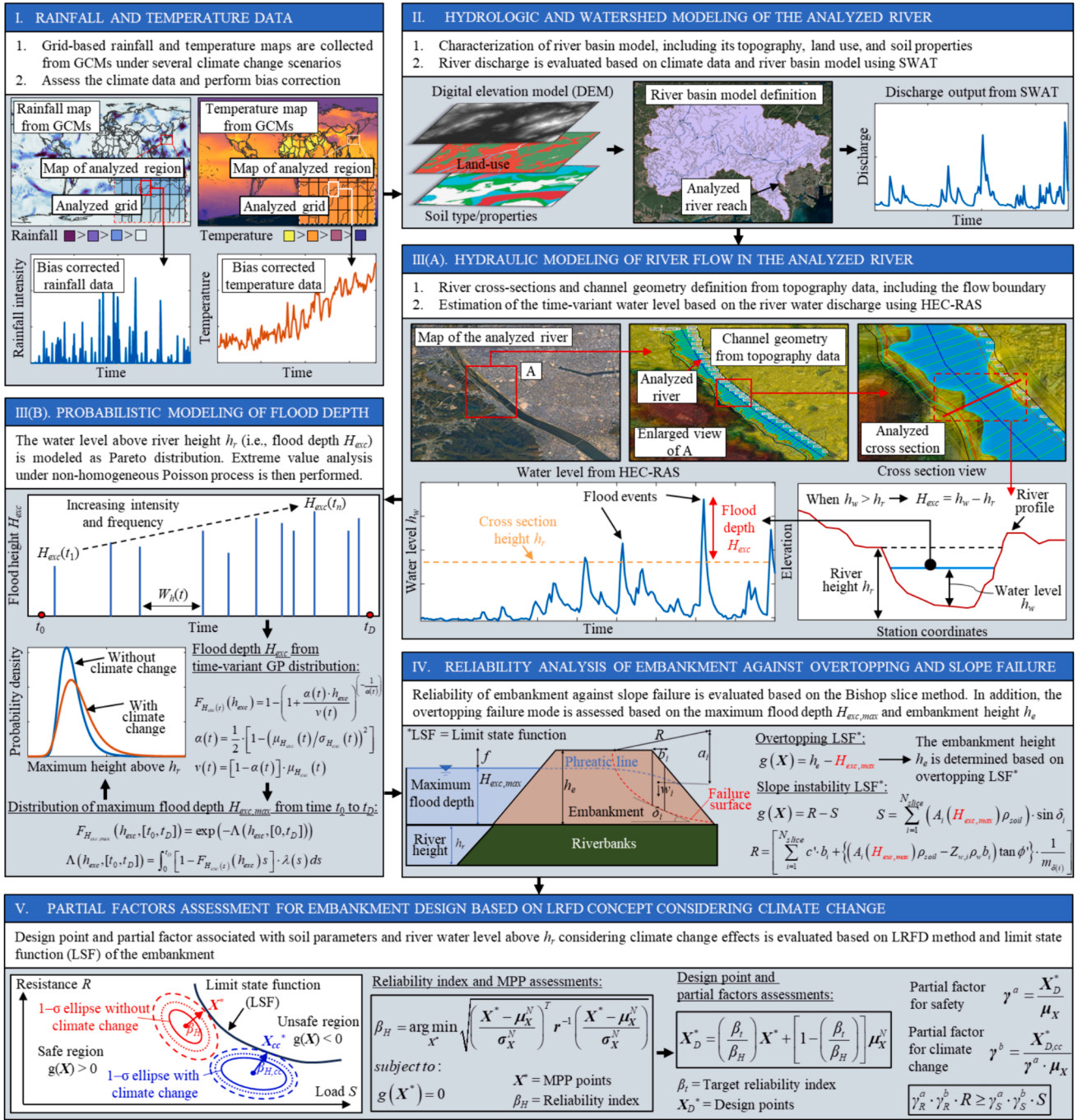


Fig. 1. Procedure to estimate the partial factors for embankment design considering non-stationary climate change effects based on the LRFD concept.

$$\gamma_R^a \cdot R_k \geq \gamma_S^a \cdot S_k$$

(1)

where R_k and S_k are the characteristic values associated with structural resistance R and load actions S associated with maximum flood depth from the present time t_0 to an analyzed time t_D , and γ_R^a and γ_S^a are the reduction and amplification factors related to R and S , respectively, which can be expressed as:

$$\gamma_R^a = \frac{R_D^*}{R_k}$$

(2)

$$\gamma_S^a = \frac{S_D^*}{S_k}$$

(3)

where R_D^* and S_D^* are the design point for R and S , respectively. In this study, R_k and S_k are assumed to be the mean values [60].

Furthermore, additional partial factors are necessary to transform the design points R_D^* and S_D^* into new design points $R_{D,cc}^*$ and $S_{D,cc}^*$ to

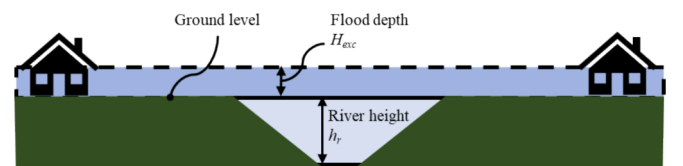


Fig. 2. Illustration of the excess water above the river height (i.e., flood depth H_{exc}) as a design variable for embankment. The flood depth H_{exc} is measured relative to the ground level.

account for climate change effects. The LRFD expression in Eq. (1) becomes:

$$\gamma_R^a \cdot \gamma_R^b \cdot R_k \geq \gamma_S^a \cdot \gamma_S^b \cdot S_k \quad (4)$$

where:

$$\gamma_R^b = \frac{R_{D,cc}^*}{R_D^*} \quad (5)$$

$$\gamma_S^b = \frac{S_{D,cc}^*}{S_D^*} \quad (6)$$

Intuitively, the partial factors γ_R^b and γ_S^b measure the extent to which the design points should be modified with respect to the original design points R_D^* and S_D^* due to climate change, as illustrated in step (5). Furthermore, climate change is anticipated to significantly impact the design load actions $S_{D,cc}$ while having a lesser effect on the resistance $R_{D,cc}$. This is because climate change directly influences the uncertainties associated with load actions. The detailed procedure to determine the partial factors for embankment considering non-stationary climate change effects is discussed in Section 4.

3. Probabilistic modeling of maximum flood depth based on hydrological and hydraulic analyses of the river

Identifying essential random variables for embankment design is imperative. One such variable is the maximum flood depth $H_{exc,max}$, which is determined through comprehensive hydrological and hydraulic analyses. The distribution of the maximum flood depth is developed herein using extreme value analysis and a non-homogeneous stochastic Poisson process, considering the time-dependent effects of climate change on the intensity and frequency of the maximum flood depth.

3.1. Bias correction and statistical downscaling of climate models

GCMs often have limited spatial resolution and may not adequately capture local climate characteristics [61]. To mitigate biases between GCM projections and local observations, the Quantile Mapping (QM) method is commonly employed by assuming similar statistical characteristics [62]. In addition, the Quantile Delta Mapping (QDM) is complementary to QM by considering the relative changes between simulated historical and future climates in GCMs. To estimate the bias-corrected GCM data, the CDF associated with the simulated historical and future projection data from GCMs (hereafter denoted as $F_{s,h,m}$ and $F_{s,p,m}$, respectively) and the observed historical data from the local climate station, indicated as $F_{o,h,m}$, are estimated for a specific month m . The bias-corrected GCM data according to the QM method can be calculated as [61]:

$$\hat{\theta}_{s,p}(t) = F_{o,h,m}^{-1} \{ F_{s,p,m} [\theta_{s,p}(t)] \} \quad (7)$$

where $\hat{\theta}_{s,p}(t)$ denote the bias-corrected climate variable (e.g., rainfall or temperature) associated with future projections from GCMs at time t , where t is defined as a date vector, expressed as $t = \{t_{year}, t_{month}, t_{day}\}$, and $\theta_{s,p}$ is the original future projection climate variable from GCMs. t_{month} in the date vector t equals m .

Moreover, the relative changes Δ_s between the historical and future data of climate variables in the GCM output can be quantified as:

$$\Delta_s(t) = \frac{\theta_{s,p}(t)}{F_{s,h,m}^{-1} \{ F_{s,p,m} [\theta_{s,p}(t)] \}} \quad (8)$$

Based on Eqs. (7) and (8), the bias-corrected GCM according to the QDM method can be expressed as:

$$\hat{\theta}_{s,p}(t) = F_{o,h,m}^{-1} \{ F_{s,p,m} [\theta_{s,p}(t)] \} \cdot \Delta_s(t) = \theta_{s,p}(t) \cdot \frac{F_{o,h,m}^{-1} \{ F_{s,p,m} [\theta_{s,p}(t)] \}}{F_{s,h,m}^{-1} \{ F_{s,p,m} [\theta_{s,p}(t)] \}} \quad (9)$$

The bias-corrected GCM climate variables obtained based on the QDM method are utilized for hydrological modeling and the evaluation of time-variant river discharge using SWAT.

3.2. Hydrological modeling and assessment of the analyzed river

The spatial information of land cover, soil type, and DEM data are employed as input to SWAT to generate unique combinations of homogeneous physical elements known as HRU [63]. In this study, the 30-m DEM data are collected from the Shuttle Radar Topography Mission (SRTM) [64], while the land cover and soil data are collected from the ESRI Sentinel-2 dataset [65] and the Digital Soil Map of the World dataset provided by the Food and Agriculture Organization (FAO) of the United Nations [66], respectively.

The hydrological cycle within the delineated watershed in the SWAT model is evaluated according to the soil water balance equation (SWBE) that involves precipitation, surface runoff, evapotranspiration, seepage, and return flow [56,63]. Based on the SWBE, SWAT routed the flows in different HRUs to the main outlet of the watershed. In this study, the surface runoff is estimated according to the curve number of the Soil Conservation Service (SCS) [67], and the routing process is assumed to follow the Muskingum routing models [68]. The details can be found in Neitsch et al. [56].

Evapotranspiration plays a significant role in SWBE and is strongly related to several climatic variables, including solar radiation, air temperature, wind speed, and relative humidity. In addition, climate change is expected to impact rainfall intensity globally, increasing it in high-latitude regions and decreasing it in low-latitude areas [69]. Conversely, the temperature is expected to increase globally. However, the simultaneous increase in rainfall intensity and temperature does not necessarily imply future increases in river discharges. Minaei and Irannezhad [70] highlighted that, despite rising precipitation and temperature in northeast Iran from 1953 to 2013, river discharge decreased due to higher temperatures that accelerated evapotranspiration. Therefore, the future trajectory of river discharge depends on the severity of climate change at the specific location under analysis.

3.3. Hydraulic modeling and assessment of the analyzed river

HEC-RAS is commonly used for riverine flooding assessments, particularly in examining river streamflow characteristics during high-flow events [71–73]. The DEM data and the river discharge are used as the input for the hydraulic analysis. In addition, the one-dimensional unsteady flow hydrodynamics model is adopted herein to estimate the river water depth based on the time-variant river discharge. The detailed description of the procedure for hydraulic modeling using HEC-RAS is beyond the scope of this study and can be found elsewhere [74,75].

The time-variant river water depth relative to the riverbed can be obtained for the non-stationary probabilistic modeling of the maximum flood depth. This probabilistic model is of paramount importance for developing an LRFD-based embankment design that considers the effects of climate change.

3.4. Probabilistic distribution of the maximum excess water height

The excess water level above the river cross-sectional height (i.e., flood depth H_{exc}), as indicated in Fig. 2, is used as the primary variable for embankment design. Flooding events are often modeled with the Poisson process [76,77]. Due to climate change effects, the intensity and frequency of maximum flood depths become a non-stationary Poisson process. In addition, the excess of water H_{exc} above the river height h_r can be approximated using the POT method [78]. A combination of the extreme value analysis using the POT method and the non-homogeneous Poisson process is developed herein to consider climate change effects.

In the first step, the intensity of the river water level exceeding the river height H_{exc} and the corresponding arrival times of these events are

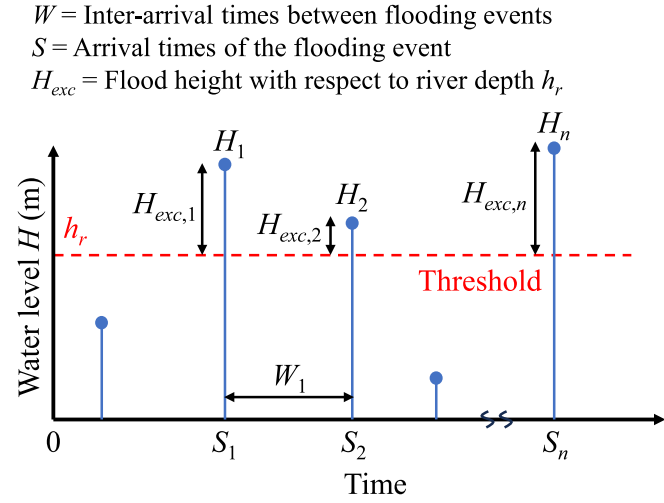


Fig. 3. Schematic illustration of the flooding events as a renewal process.

recorded based on the time-variant river water level from the hydraulic assessment, as illustrated in Fig. 3. From the statistics of the inter-arrival time between flooding events, the mean recurrence rate λ can be estimated. Let the inter-arrival and arrival times of the flooding events be denoted as W and S , respectively. The mean and standard deviation associated with the random variable H_{exc} and the mean recurrence rate of the flood events λ are assumed to increase linearly with time t , such that:

$$\mu_{H_{exc}}(t) = \mu_{H_{exc},0} + m_{H_{exc},\mu} \cdot t \quad (10)$$

$$\sigma_{H_{exc}}(t) = \sigma_{H_{exc},0} + m_{H_{exc},\sigma} \cdot t \quad (11)$$

$$\lambda(t) = \lambda_0 + m_\lambda \cdot t \quad (12)$$

where $\mu_{H_{exc}}(t)$ and $\sigma_{H_{exc}}(t)$ are the time-variant mean and standard deviation of the flood depth H_{exc} (m), respectively, $\lambda(t)$ is the time-variant mean recurrence rate of the flooding events (event per unit of time), $\mu_{H_{exc},0}$ and $\sigma_{H_{exc},0}$ are the mean and standard deviation of the flood depth at present time t_0 , respectively, λ_0 is the mean recurrence rate of the flooding events at present time t_0 , $m_{H_{exc},\mu}$ and $m_{H_{exc},\sigma}$ are the rate of change per amount of time t associated with the mean and standard deviation of the flood depth H_{exc} (meter per amount of time), respectively, and m_λ is the rate of change per amount of time t for the mean recurrence rate λ . These time-variant parameters are statistically

obtained from the output of hydraulic assessment illustrated in Fig. 3 using time-series analysis such as moving average (MA) and moving standard deviation (MSD) methods [79,80].

The Generalized Pareto (GP) distribution is often utilized for modeling exceedances due to its adaptability in capturing the tail behavior of the distribution [81–83]. However, the GP distribution becomes time-dependent due to the effects of non-stationary climate change. The CDF of the time-variant GP distribution and the corresponding statistical parameters can be expressed as:

$$F_{H_{exc}(t)}(h_{exc}) = \begin{cases} 1 - \left(1 + \frac{\alpha(t) \cdot h_{exc}}{\nu(t)}\right)^{\left(-\frac{1}{\alpha(t)}\right)} & \text{for } \alpha \neq 0 \\ 1 - \exp\left(-\frac{h_{exc}}{\nu(t)}\right) & \text{for } \alpha = 0 \end{cases} \quad (13)$$

$$\alpha(t) = \frac{1}{2} \left[1 - \left(\frac{\mu_{H_{exc}}(t)}{\sigma_{H_{exc}}(t)} \right)^2 \right] \quad (14)$$

$$\nu(t) = [1 - \alpha(t)] \cdot \mu_{H_{exc}}(t) \quad (15)$$

where $F_{H_{exc}(t)}(h_{exc})$ describes the probability of H_{exc} having a value less than h_{exc} at time t , $\alpha(t)$ and $\nu(t)$ are the time-variant shape and scale parameters associated with the GP distribution, respectively. Assuming that the flood events follow a non-homogeneous Poisson process, the CDF of the maximum flood depth can be evaluated as [84]:

$$F_{H_{exc,max}(t_D)}(h_{exc}) = \exp\left\{-\int_{t_0}^{t_D} [1 - F_{H_{exc}(s)}(h_{exc})] \cdot \lambda(s) ds\right\} \quad (16)$$

$$f_{H_{exc,max}(t_D)}(h_{exc}) = \frac{dF_{H_{exc,max}(t_D)}(h_{exc})}{dh_{exc}} \quad (17)$$

where $F_{H_{exc,max}(t_D)}(h_{exc})$ and $f_{H_{exc,max}(t_D)}(h_{exc})$ are the CDF and the probability density function (PDF) of maximum flood depth from present time t_0 to an analyzed time t_D , respectively, and $F_{H_{exc}(s)}(h_{exc})$ is the time-variant GP distribution with statistical parameters according to Eqs. (14) and (15). Eq. (16) and (17) can be solved straightforwardly without MCS.

4. LRFD-based embankment design considering non-stationary climate change effects

The distribution of maximum flood depth described in the previous section is of paramount importance since it governs the load actions of embankments associated with the overtopping and slope instability

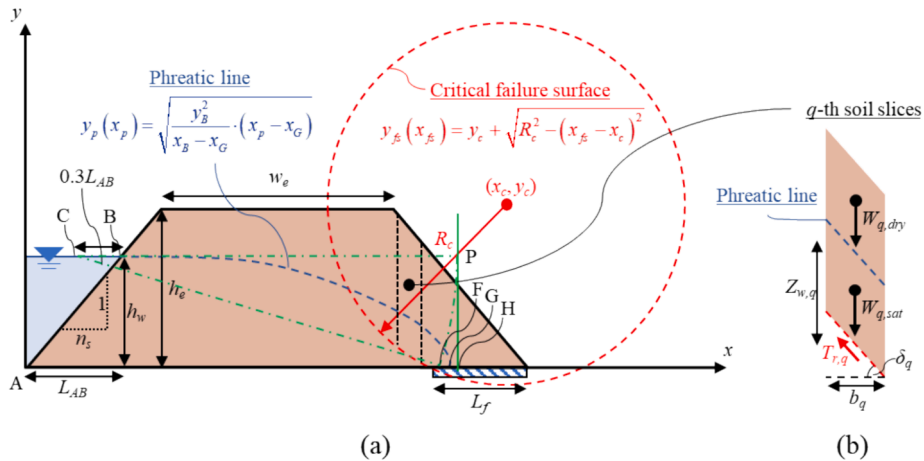


Fig. 4. (a) Geometry definition for the slope stability assessment of embankment, and (b) geometry and weight forces acting on the q -th soil slices during slope stability assessment.

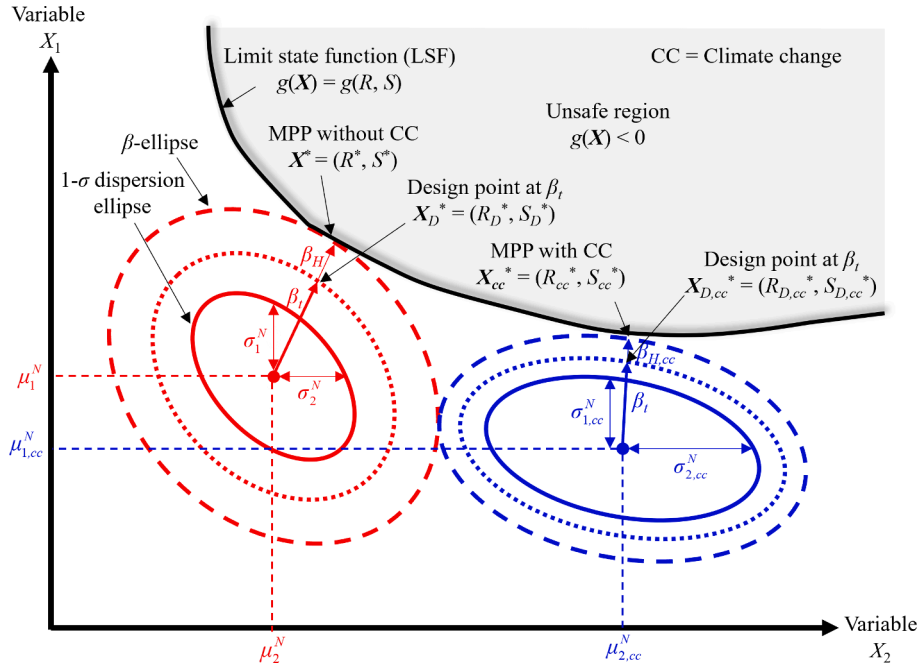


Fig. 5. Equivalent normal 1- σ dispersion ellipsoids, most probable failure points, and design points associated with embankment design parameters for correlated random variables. (Note that red lines indicate the original reliability index without climate change effects; and blue lines indicate the reliability index when considering climate change effects). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

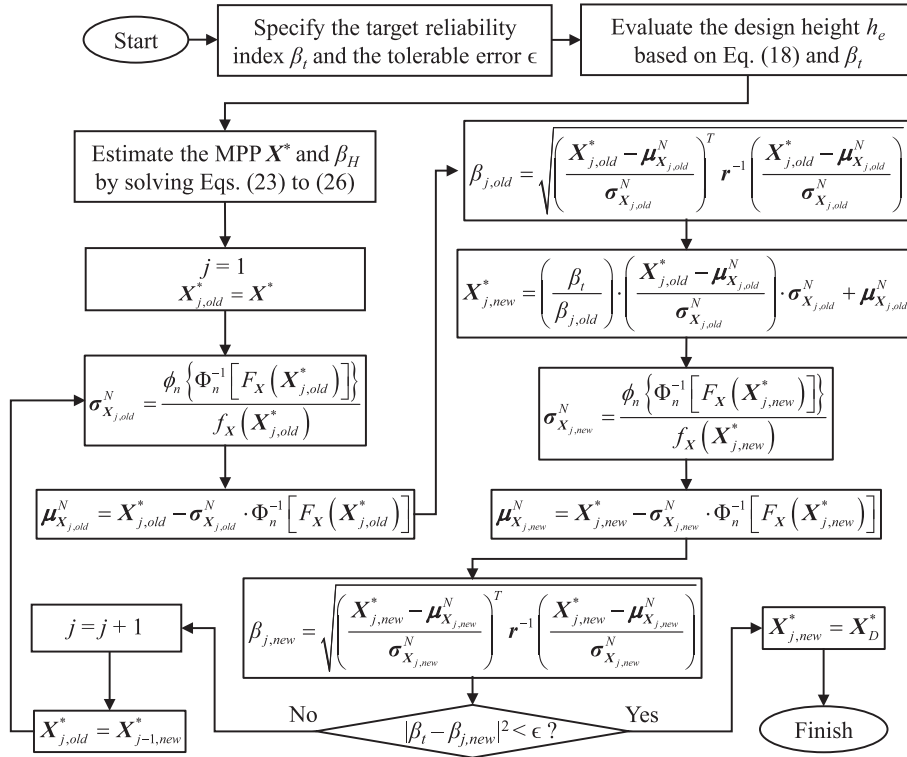


Fig. 6. Iterative procedure to estimate the design point X_D^* .

failure modes. During the reliability assessment of an embankment, the maximum flood depth distribution given in Eqs. (16) and (17) are utilized to estimate the statistical parameters associated with equivalent normal distribution for random variable $H_{exc,max}$. Along with the uncertainties associated with soil parameters, the LRFD-based design methodology considering non-stationary climate change effects is developed herein.

4.1. Limit state functions for embankments against overtopping and slope instability

Based on previous disasters in Japan, the existing embankments were reported to be vulnerable to slope failure and water overtopping [85,86]. To introduce the proposed method, linear and nonlinear failure modes associated with embankment overtopping and slope instability

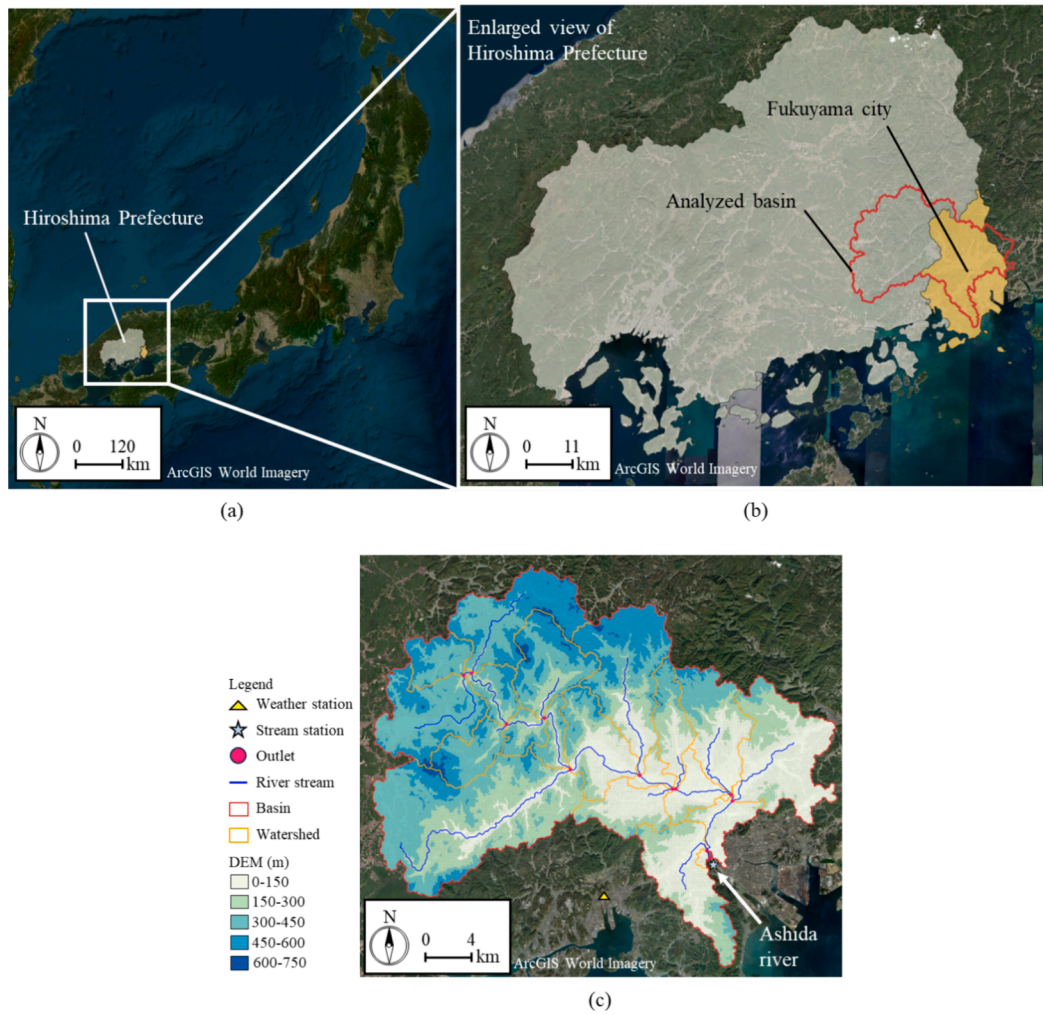


Fig. 7. (a) Hiroshima Prefecture location with respect to Japan, (b) Fukuyama city and the analyzed basin, and (c) the analyzed river watershed in Fukuyama city and the associated DEM.

are presented herein. While piping failure is not explicitly addressed [87], the proposed method remains applicable to this failure mode. It is assumed that the embankment is constructed utilizing soil-based materials, following the prevalent practice of embankment construction in Japan [88–90].

An overtopping event in the context of an embankment and other flood defence systems refers to a situation where the water level exceeds the embankment design height. Therefore, the limit state function for embankment overtopping can be expressed as:

$$g_o(\mathbf{X}_o) = h_e - H_{exc,max} \quad (18)$$

where $g_o(\mathbf{X}_o)$ denotes the limit state function for overtopping with $\mathbf{X}_o$ as the associated random variables, and h_e is the designated embankment height. According to Eq. (18), $\mathbf{X}_o = \{H_{exc,max}\}$.

On the other hand, the definition of the embankment geometry for slope stability assessment is shown in Fig. 4. The phreatic line is substantially influenced by the maximum flood depth $H_{exc,max}$ and the horizontal drain filter length L_f . The horizontal drain filter length L_f can be determined within a range defined by its minimum and maximum and can be found in Chahar [91]. In this study, the horizontal drain filter length is set as 1.3 times its minimum value to ensure that the phreatic line consistently remains positioned behind the face of the downstream slope. Once the horizontal drain filter length L_f is estimated, the phreatic line geometry is assumed to follow a parabolic equation, as shown in

Fig. 4(a). In addition, the abscissa of the vertex point x_G can be evaluated according to the Casagrande graphical method [92].

The failure surface, modeled as a circle in Fig. 4(a), involves identifying the critical surface with the lowest safety factor (i.e., the ratio between soil shear strength and gravitational force acting on the slope). However, this iterative approach can be impractical and computationally intensive. Therefore, a response surface is developed using cubic interpolations by sampling several embankment height and width combinations. The critical failure surface parameters associated with the center coordinates of the critical failure surfaces (i.e., x_c and y_c) and the corresponding radius R_c are evaluated using the widely employed software GEOSTUDIO [93]. It should be noted that the effective cohesion and the friction angle have negligible effects on the geometry of the critical failure surface.

Once all the geometries associated with the phreatic line, critical failure surface, and horizontal drain filter length are defined, the limit state function for embankment slope instability can be expressed by the Bishop slice method [94] as:

$$g_s(\mathbf{X}_s) = R - S \quad (19)$$

$$R = \sum_{q=1}^{N_{slice}(h_e)} c' \cdot b_q + \left\{ (A_{q,dry} \rho_{dry} + A_{q,sat} \rho_{sat} - Z_{w,q} \rho_w b_q) \tan \phi' \right\} \cdot \frac{1}{m_{\phi(q)}} \quad (20)$$

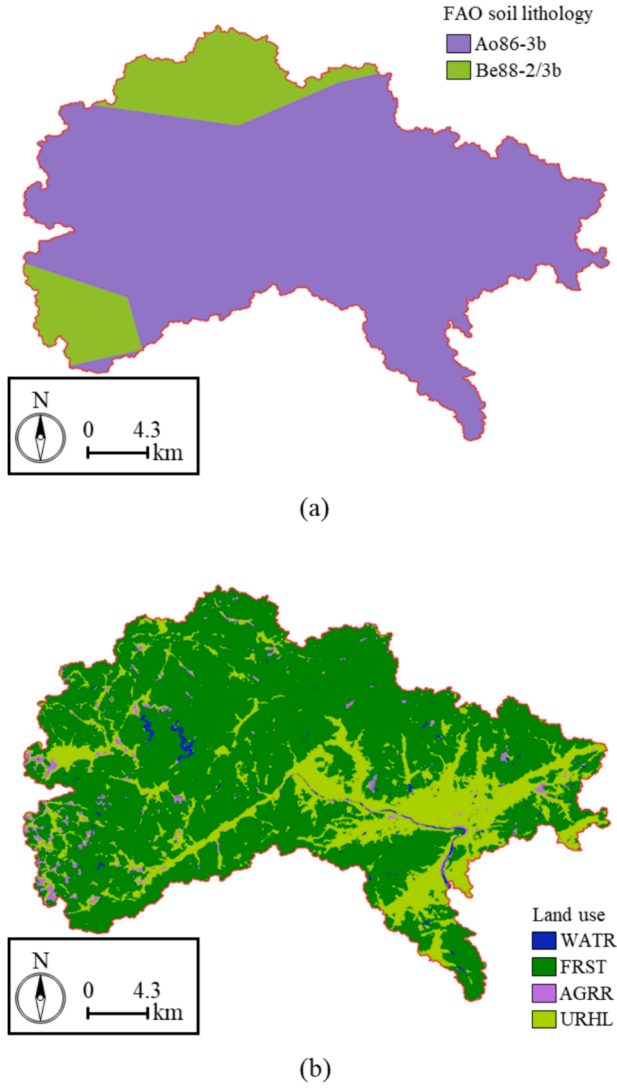


Fig. 8. Geospatial information for hydrological modeling: (a) FAO soil lithology, and (b) land use. WATR = water bodies, FRST = forest (mixed), AGRR = agricultural land (row crops), and URHL = residential (high-density).

$$S = \sum_{q=1}^{N_{\text{slice}}(h_e)} (A_{q,\text{dry}}\rho_{\text{dry}} + A_{q,\text{sat}}\rho_{\text{sat}}) \cdot \sin\delta_q \quad (21)$$

$$m_{\delta(p)} = \cos\delta_p + \sin\delta_p \cdot \tan\phi' \quad (22)$$

where $g_s(\mathbf{X}_s)$ is the limit state function for slope instability with $\mathbf{X}_s$ as the associated set of random variables, $N_{\text{slice}}(h_e)$ is the total number of slices as a function of embankment height h_e , R and S denote resisting force and driving force of the soil slope, respectively, c' is the effective soil cohesion (kPa), b_q is the width of the q -th soil slices (m), $A_{q,\text{dry}}$ and $A_{q,\text{sat}}$ are the cross-sectional area of q -th soil above and below the phreatic line, respectively (m^2), ρ_{dry} and ρ_{sat} are the unit weight of the soil during dry and saturated conditions, respectively (kN/m^3), $Z_{w,q}$ is the depth of water table relative to the critical failure surface (m), which can be approximated as $y_{p,q} - y_{fs,q}$, ρ_w is the unit weight of water (kN/m^3), ϕ' is the effective friction angle ($^\circ$), and δ_q is the failure surface angle of q -th soil relative to a horizontal line (see Fig. 4(b)). Although $H_{\text{exc,max}}$ is not explicitly described in Eqs. (19) to (22), it influences the geometry of the phreatic line, implicitly affecting the variables $A_{q,\text{dry}}$ and $A_{q,\text{sat}}$. As the phreatic line rises, the submerged soil area $A_{q,\text{sat}}$ for each slice expands, consequently amplifying the gravitational weight $W_{q,\text{sat}}$ of the soil. This

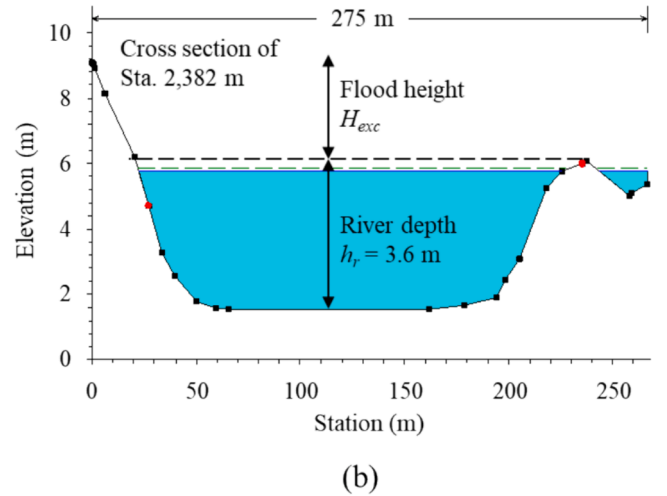
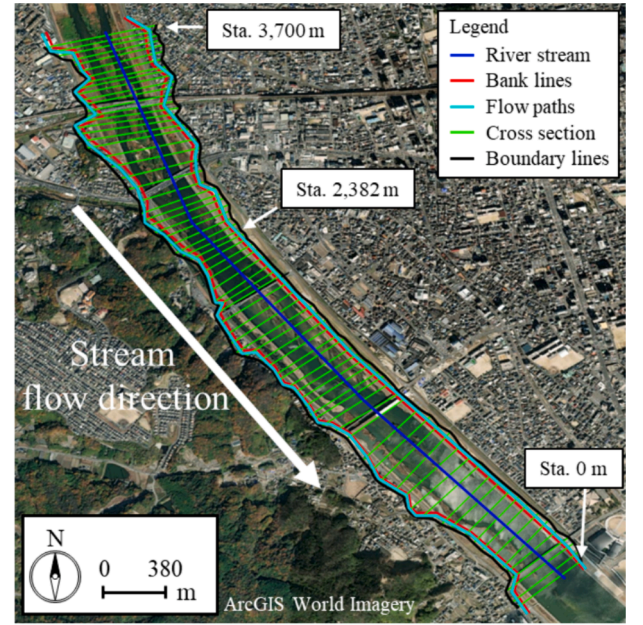


Fig. 9. (a) Hydraulic model and river geometry definition of Ashida River based on HEC-RAS, and (b) cross-section of river in Sta. 2,382 m.

leads to a simultaneous increase in both the resistance R and load action S .

The reliability of structures can be evaluated through MCS. However, this approach involves significantly intricate computations. Instead, this paper aims to present a methodology to determine partial factors to ensure the target reliability without intricate computations and to facilitate the consideration of climate change effects for practical designers. The most probable points (MPP) and design points of the random variables are evaluated to determine the partial factors, which are discussed in the following subsection.

4.2. Partial factor estimation considering climate change effects

To determine the partial factor considering climate change effects, the embankment height should be designed with sufficient reliability to prevent overtopping and slope instability failure. Since the POT method is adopted herein instead of the block maxima approach (see [supplementary material](#)), the embankment reliability index and the associated

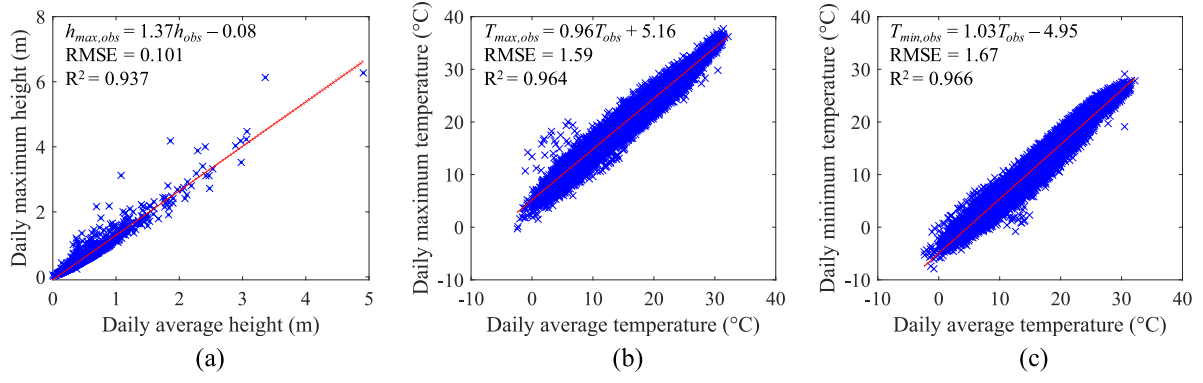


Fig. 10. (a) Relationship between the daily average water height and daily maximum water height, (b) relationship between the daily average temperature and daily maximum temperature, (c) and relationship between the daily average temperature and daily minimum temperature.

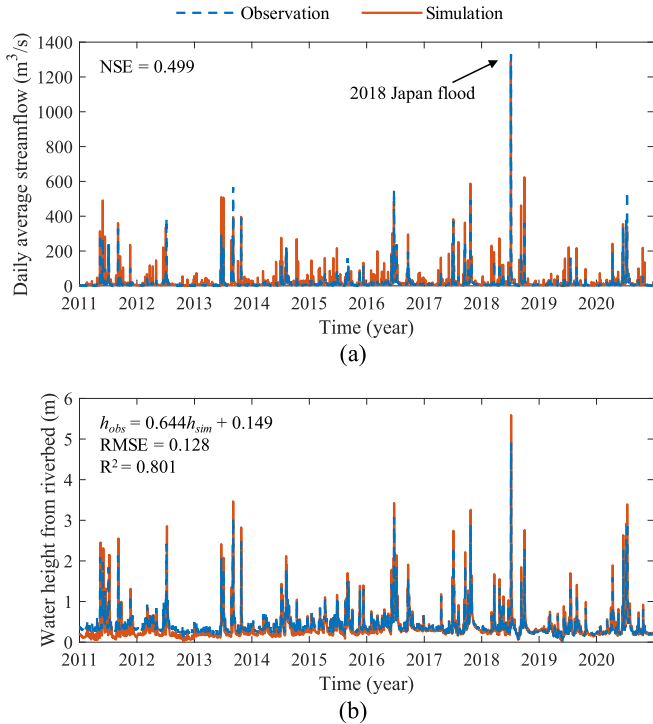


Fig. 11. Verification results for (a) daily average streamflow from hydrological assessment and (b) water height from hydraulic assessment.

target reliability index refer to the reliability from time t_0 to time t_D . Furthermore, due to the intricate expression in Eqs. (19) to (22), it becomes increasingly complex to employ the FORM method since it necessitates the transformation to the standard normal space and involves derivating the limit state function with respect to each random variable. Although the method chosen in this paper does not guarantee correctness and convergence in general, it worked fine in this case. Therefore, a more suitable way is to apply the ellipsoid approach introduced by Low & Tang [95]. This approach aims to find the minimum ellipsoid that touches the failure surface, as illustrated in Fig. 5, by satisfying the following constrained nonlinear optimization problem:

$$\beta_H = \underset{\mathbf{X}^*}{\operatorname{argmin}} \sqrt{\left(\frac{\mathbf{X}^* - \boldsymbol{\mu}_X^N}{\boldsymbol{\sigma}_X^N} \right)^T \mathbf{r}^{-1} \left(\frac{\mathbf{X}^* - \boldsymbol{\mu}_X^N}{\boldsymbol{\sigma}_X^N} \right)} \quad (23)$$

subject to:

$$g(\mathbf{X}^*) = 0 \quad (24)$$

$$\boldsymbol{\mu}_X^N = \mathbf{X}^* - \boldsymbol{\sigma}_X^N \cdot \Phi_n^{-1} [F_X(\mathbf{X}^*)] \quad (25)$$

$$\boldsymbol{\sigma}_X^N = \frac{\phi_n \{ \Phi_n^{-1} [F_X(\mathbf{X}^*)] \}}{f_X(\mathbf{X}^*)} \quad (26)$$

where β_H is the Hasofer-Lind reliability index without climate change effects, as indicated in Fig. 5, $\mathbf{X}^*$ is the set of MPP of the corresponding failure mode, $\boldsymbol{\mu}_X^N$ and $\boldsymbol{\sigma}_X^N$ are the set of normalized mean and standard deviation, respectively, following the concept introduced by Rackwitz & Fiessler [96], ϕ_n and Φ_n are the PDF and CDF of standard normal distribution, respectively, and $\mathbf{r}$ is the correlation matrix. Since the limit state function in Eqs. (19) to (22) is significantly complex due to the dependence of the phreatic line geometry on the maximum flood depth $H_{exc,max}$, the sequential quadratic programming (SQP) method is adopted to solve the nonlinear constrained optimization in Eqs. (23) to (26). The details of the SQP algorithm are described thoroughly in Boggs & Tolle [97].

To achieve a predetermined level of safety, the design point of each variable is evaluated based on the target reliability β_t with $\beta_t < \beta_H$. In addition, to assess the value of the design point $\mathbf{X}_D^*$, it is necessary to obtain the original reliability index β_H and the corresponding MPP $\mathbf{X}^*$ by solving Eqs. (23) to (26). It should be noted that the limit state functions in Eqs. (19) to (22) associated with the constraint in Eq. (24) are dependent on the embankment height h_e , which is designed according to Eq. (18) and reliability target β_t . The iterative procedure to estimate the design point $\mathbf{X}_D^*$ is shown in Fig. 6 and described as follows:

- (1) Specify the target reliability index β_t and the tolerable error ϵ .
- (2) Determine the embankment design height h_e according to the limit state function in Eq. (18) and target reliability index β_t .
- (3) Evaluate the reliability index β_H and MPP point $\mathbf{X}^*$ by solving Eqs. (23) to (26).
- (4) In the first iteration (i.e., $j = 1$), define $\mathbf{X}^*$ as $\mathbf{X}_{j,old}^*$.
- (5) Evaluate the normalized mean $\boldsymbol{\mu}_{X_{j,old}}^N$ and standard deviation $\boldsymbol{\sigma}_{X_{j,old}}^N$ for the j -th iteration based on $\mathbf{X}_{j,old}^*$ as follows:

$$\boldsymbol{\mu}_{X_{j,old}}^N = \mathbf{X}_{j,old}^* - \boldsymbol{\sigma}_{X_{j,old}}^N \cdot \Phi_n^{-1} [F_X(\mathbf{X}_{j,old}^*)] \quad (27)$$

$$\boldsymbol{\sigma}_{X_{j,old}}^N = \frac{\phi_n \{ \Phi_n^{-1} [F_X(\mathbf{X}_{j,old}^*)] \}}{f_X(\mathbf{X}_{j,old}^*)} \quad (28)$$

- (6) Assess the reliability index $\beta_{j,old}$ for the j -th iteration according to the following expression

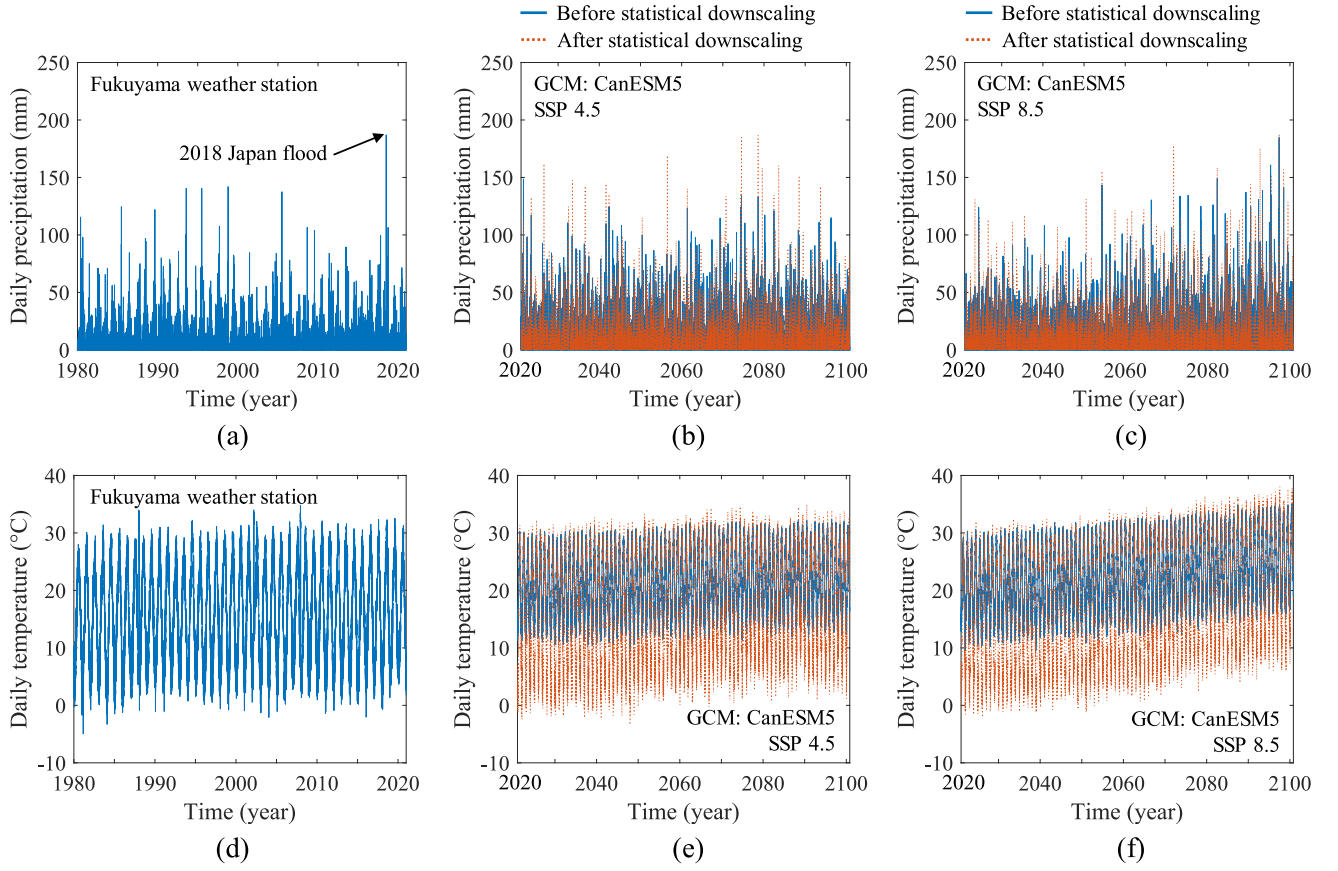


Fig. 12. (a) Observed daily precipitation from Fukuyama weather station, statistical downscaling of daily precipitation of CanESM5 GCM for (b) SSP 4.5, (c) SSP 8.5, observed daily temperature from Fukuyama weather station, statistical downscaling of the daily temperature of CanESM5 GCM for (b) SSP 4.5, (c) SSP 8.5.

$$\beta_{j,old} = \sqrt{\left(\frac{X_{j,old}^* - \mu_{X_{j,old}}^N}{\sigma_{X_{j,old}}^N} \right)^T r^{-1} \left(\frac{X_{j,old}^* - \mu_{X_{j,old}}^N}{\sigma_{X_{j,old}}^N} \right)} \quad (29)$$

(7) A reduction factor of $\psi_j = (\beta_t / \beta_{j,old})$ is introduced and employed to estimate the new MPP $X_{j,new}^*$ and the associated normalized mean $\mu_{X_{j,old}}^N$ and standard deviation $\sigma_{X_{j,old}}^N$ as follows

$$\begin{aligned} X_{j,new}^* &= \psi_j \cdot \left(\frac{X_{j,old}^* - \mu_{X_{j,old}}^N}{\sigma_{X_{j,old}}^N} \right) \cdot \sigma_{X_{j,old}}^N + \mu_{X_{j,old}}^N \\ &= \left(\frac{\beta_t}{\beta_{j,old}} \right) \cdot \left(\frac{X_{j,old}^* - \mu_{X_{j,old}}^N}{\sigma_{X_{j,old}}^N} \right) \cdot \sigma_{X_{j,old}}^N + \mu_{X_{j,old}}^N \end{aligned} \quad (30)$$

$$\mu_{X_{j,new}}^N = X_{j,new}^* - \sigma_{X_{j,new}}^N \cdot \Phi_n^{-1} \left[F_X(X_{j,new}^*) \right] \quad (31)$$

$$\sigma_{X_{j,new}}^N = \frac{\phi_n \left\{ \Phi_n^{-1} \left[F_X(X_{j,new}^*) \right] \right\}}{f_X(X_{j,new}^*)} \quad (32)$$

(8) Evaluate the new reliability index $\beta_{j,new}$ in the same manner as Eq. (29)

$$\beta_{j,new} = \sqrt{\left(\frac{X_{j,new}^* - \mu_{X_{j,new}}^N}{\sigma_{X_{j,new}}^N} \right)^T r^{-1} \left(\frac{X_{j,new}^* - \mu_{X_{j,new}}^N}{\sigma_{X_{j,new}}^N} \right)} \quad (33)$$

If the condition $|\beta_t - \beta_{j,new}|^2 < \epsilon$ is satisfied, with ϵ as the predefined tolerable error, $X_{j,new}^*$ is assigned as the design point X_D^* . Otherwise,

proceed to the next iteration (i.e., $j = j + 1$) and repeat steps (2) to (5) with $X_{j,old}^* = X_{j-1,new}^*$ until the condition $|\beta_t - \beta_{j,new}|^2 < \epsilon$ is satisfied.

Due to the impacts of climate change, numerous studies have revealed that the capacity of structures is rapidly diminishing (e.g., concrete corrosion under climate change [98]) and that both the intensity and frequency of load actions are increasing [99]. Let X_1 and X_2 be defined as the random variables that are strongly correlated to the resistance R and load S , respectively (e.g., in the context of structural engineering X_1 could be the yield strength of steel and X_2 is the chloride concentration), with $\{X_1, X_2\} \subseteq \mathbf{X}$. As a result, the normalized mean of structural resistance μ_1^N in Fig. 5 is projected to decrease and become $\mu_{1,cc}^N$, while the normalized load actions μ_2^N increases and becomes $\mu_{2,cc}^N$ due to the climate change effects. In addition, since climate change is deemed a non-stationary process [46,47], the normalized standard deviation of structural resistance and load actions become $\sigma_{1,cc}^N$ and $\sigma_{2,cc}^N$, respectively. These alterations would imply a shift in the center points and enlarge the radius of the equivalent normal 1- σ dispersion ellipsoids, as illustrated in Fig. 5.

The overall process of estimating the partial factors for LRFD-based design considering climate change effects is summarised as follows:

1. Determine the target reliability index β_t , characteristic values of random variables $\mathbf{X}_k$, and the associated PDF and CDF, and embankment height h_e based on Eq. (18) and β_t .
2. Estimate the reliability index β_H and the associated MPP $\mathbf{X}^*$ by solving Eqs. (23) to (26).
- 2'. When considering climate change effects, the reliability index and the associated MPP are assessed and denoted as $\beta_{H,cc}$ and X_{cc}^* , respectively, with $\beta_{H,cc} < \beta_H$.

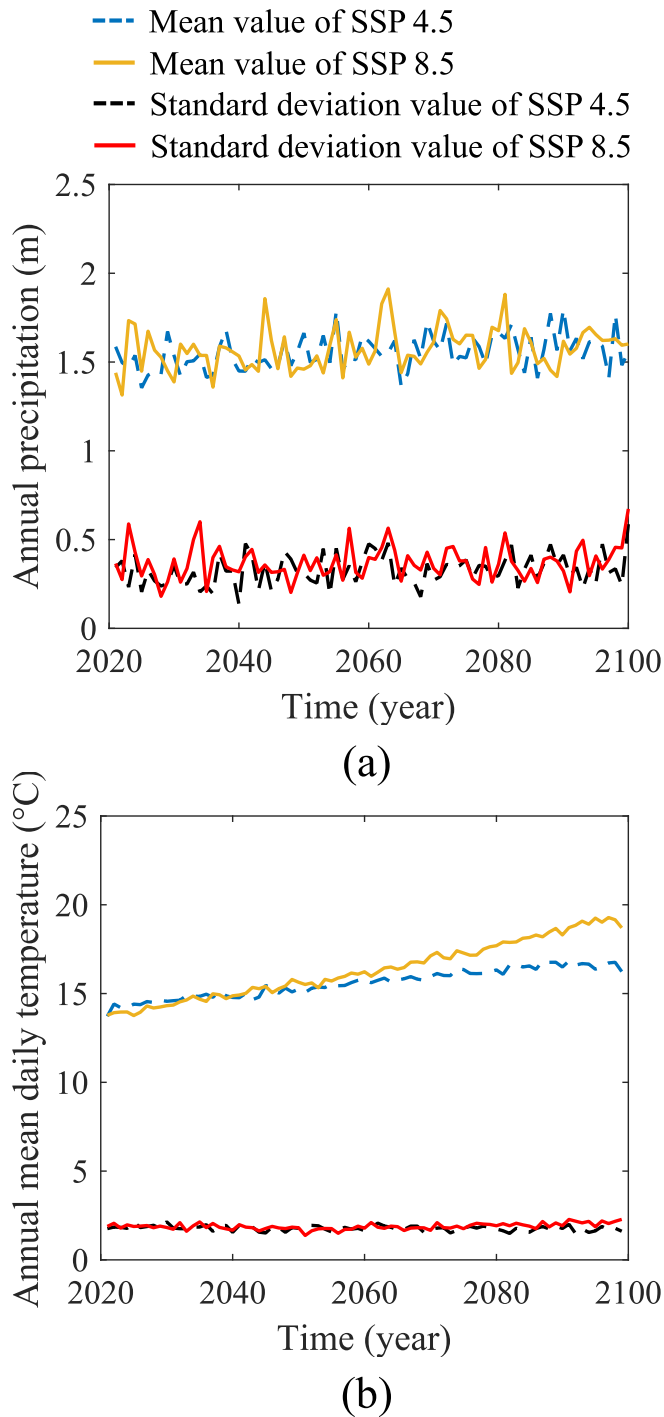


Fig. 13. Mean and standard deviation values from all GCMs for (a) annual precipitation and (b) annual mean daily temperature.

3. Evaluate the design points X_D^* according to the iterative procedure in Fig. 6.
- 3'. When considering climate change effects, evaluate the design points $X_{D,cc}^*$ according to the iterative procedure in Fig. 6.
4. Assess the characteristic values associated with structural resistance R_k and load actions S_k according to Eqs. (20) and (21), respectively.
5. Similar to step (4), calculate the design points associated with structural resistance R_D^* and load actions S_D^* based on X_D^* .

- 5'. Considering climate change effects, calculate the design points for structural resistance $R_{D,cc}^*$ and load actions $S_{D,cc}^*$ based on $X_{D,cc}^*$ in a similar manner as steps (4) and (5).
6. The reduction factor γ_R^a and amplification factors γ_S^a to ensure structural safety are evaluated according to Eqs. (2) and (3).
7. The reduction factor γ_R^b and amplification factors γ_S^b to consider climate change effects are evaluated according to Eqs. (5) and (6).

As the number of random variables influenced by climate change increases, the partial factors γ_R^b and γ_S^b will progressively deviate from the value of 1.

The LRFD framework introduced in this section offers a versatile approach that extends its applicability beyond embankment cases, such as bridges under scouring hazards, roof structures against wind and rain loads, and seawalls to mitigate sea-level rise. Given the non-stationary structural deterioration and increasing load actions in the future, this framework can be considered for adoption to address the challenges posed by climate change. To demonstrate the effectiveness of the proposed framework, illustrative examples are presented in the following section.

5. Illustrative examples

5.1. Analyzed river and verification of hydrological and hydraulic model

In the 2018 Japan flood, Fukuyama city's coastal areas suffered significant damage [100]. Therefore, a river embankment near the Ashida River in Fukuyama city, Japan, is designed according to the proposed LRFD framework. The city's location in Hiroshima Prefecture and the associated topographic information are shown in Fig. 7. Moreover, the spatial distribution of land use and soil types of Fukuyama city are employed to SWAT, as shown in Fig. 8. On the other hand, the hydraulic model for the Ashida River depicted in Fig. 9 is assumed to have a uniform river height ($h_r = 3.6$ m). Based on these data, the hydrological and hydraulic assessments were carried out on a daily basis. However, to provide greater accuracy of the maximum water level height and to reduce the computational demand, the relationship between the daily average and 24-hour maximum river water level is derived from Fukuyama city's stream station data (Fig. 10(a)). For SWAT input purposes, the maximum and minimum daily temperature are derived from the daily average temperature using linear relationships to eliminate the need for hourly temperature data from each GCM, as shown in Fig. 10(b) and (c).

Fig. 11 shows the verification results of the hydrological and hydraulic model developed in this study based on the observed data from weather and stream stations. According to Fig. 11(a), the SWAT results overestimated the river baseline flow and performed well in predicting the extreme river flow, especially during the 2018 Japan flood. The performance of the SWAT model can be quantified using the Nash-Sutcliffe efficiency (NSE) [101]. The NSE value for the uncalibrated SWAT model is 0.499, implying that the SWAT performance is satisfactory [102]. On the other hand, the hydraulic model performed very well in predicting the river height for both river baseline and extreme conditions, as shown in Fig. 11(b). The relationships provided in Fig. 11 (b) and 10(a) are utilized to correct the river height obtained from hydrological and hydraulic analyses considering climate change effects, as discussed in the following subsections.

5.2. Hydrological and hydraulic analyses of river considering climate change effects

Prior to hydrological and hydraulic analyses, the rainfall data and temperature for each GCM were statistically downscaled using the QDM method described in Section 3. Daily rainfall and temperature data from 8 GCMs and 4 SSP emission scenarios (i.e., SSP 2.6, 4.5, 6.0, and 8.5) are

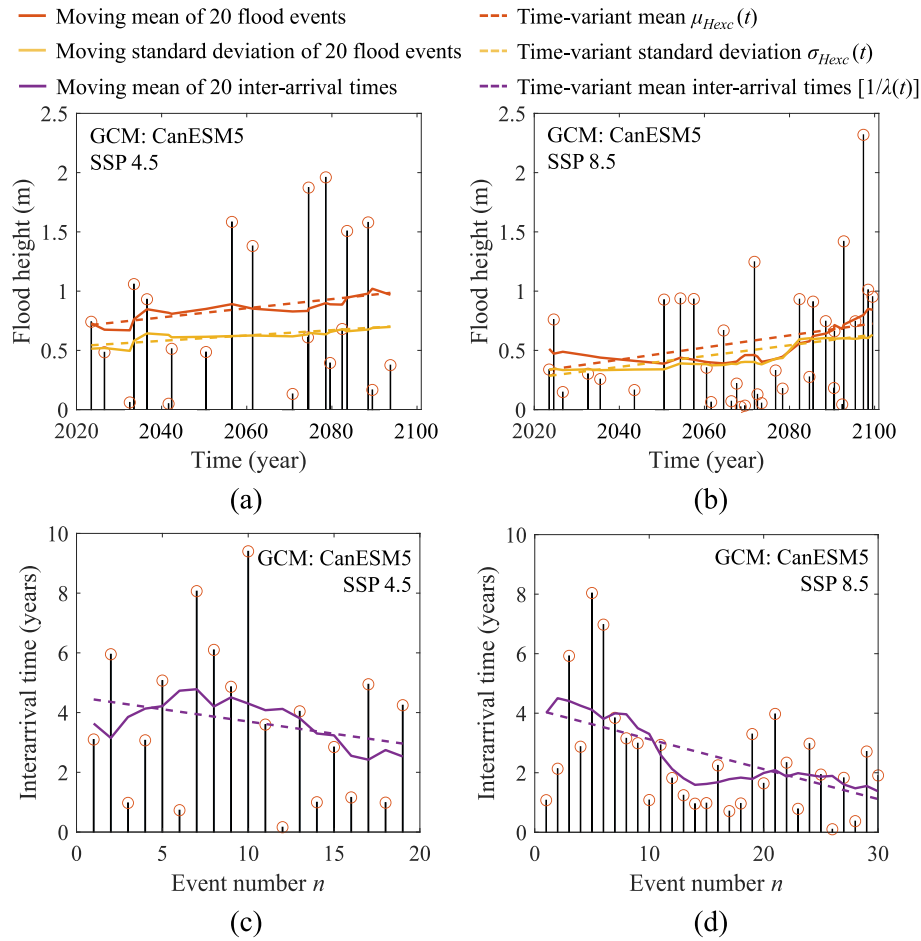


Fig. 14. Time-variant mean and standard deviation of flood depth for GCM CanESM5 under (a) SSP 4.5 and (b) SSP 8.5, and the mean inter-arrival times between flooding events for GCM CanESM5 under (c) SSP 4.5, and (d) SSP 8.5.

considered herein, spanning 2021 to 2100. Fig. 12 shows the observed and statistically downscaled precipitation and temperature data from the Fukuyama weather station, specifically for the Canadian GCM CanESM5. Daily temperature is assumed to be a normal distribution [103], while Kernel density estimation (KDE) is adopted for daily precipitation due to the poor performance of the gamma distribution for extreme values. As shown in Fig. 12, the statistical characteristics of precipitation and temperature become identical to that of the corresponding observed data, highlighting the importance of statistical downscaling.

The effects of climate change emission scenarios on precipitation and temperature were assessed by evaluating the mean and standard deviation of all GCMs under different SSPs and are presented in Fig. 13. The annual precipitation trends are relatively similar for SSP 4.5 and 8.5, with higher SSP values indicating more significant variability. Conversely, a substantial increase in temperature was projected under SSP 8.5 compared to SSP 4.5. These observations suggest that river streamflow in the Ashida River is expected to be lower under more severe climate change conditions due to the higher evapotranspiration rate. The results of hydrological and hydraulic analyses for the Ashida River are given in the [supplementary material](#).

5.3. Probabilistic assessment of the maximum flood depth

To assess the distribution of maximum flood depth considering the non-stationary climate change effects, the statistics associated with the flood depth are evaluated by subtracting the water level results from the hydraulic assessment with the river height h_r . Moreover, the time-

variant parameters associated with the mean, standard deviation, and mean recurrence rate of flood events are estimated through the MA and MSD approaches based on these statistics. For instance, Fig. 14 presents the flood depth and interarrival time between successive flood events projected by GCM CanESM5 under SSP 4.5 and 8.5. The MA and MSD were evaluated with 20 consecutive data points in each average calculation to obtain a relatively smooth moving average, as shown in Fig. 14. The MA and MSD data are then fitted into linear trends to simplify the time-varying statistical parameters. According to the CanESM5 GCM model, it is projected that the mean and standard deviation for flood depth will be more pronounced under SSP 4.5 compared to SSP 8.5. In contrast, a reduced mean inter-arrival time is observed for SSP 8.5, indicating that, despite the lower flood intensity in SSP 8.5, an upsurge in the frequency of flood events is anticipated in the future. It is noteworthy that the mean recurrence rate λ can be determined as the reciprocal of the inter-arrival time.

Fig. 15 shows the distribution of maximum flood depth evaluated according to Eqs. (16) and (17) for each GCM and SSP. The probability distribution of the maximum flood depth for each GCM tends to exhibit higher values in response to less severe climate change scenarios. The average PDF indicated in Fig. 15(a) to (e) was obtained by combining multiple individual PDFs associated with each GCM, assuming equal weights for each distribution. The expected value of the average PDF among GCMs for SSP 2.6, 4.5, 6.0, and 8.5 were evaluated to be 3.20, 3.09, 2.94, and 2.70 m, respectively. Similarly, the variability of the average PDF is lower for a higher SSP. This trend is attributable to similar precipitation patterns among SSPs and lower temperature that decelerates the evapotranspiration rate.

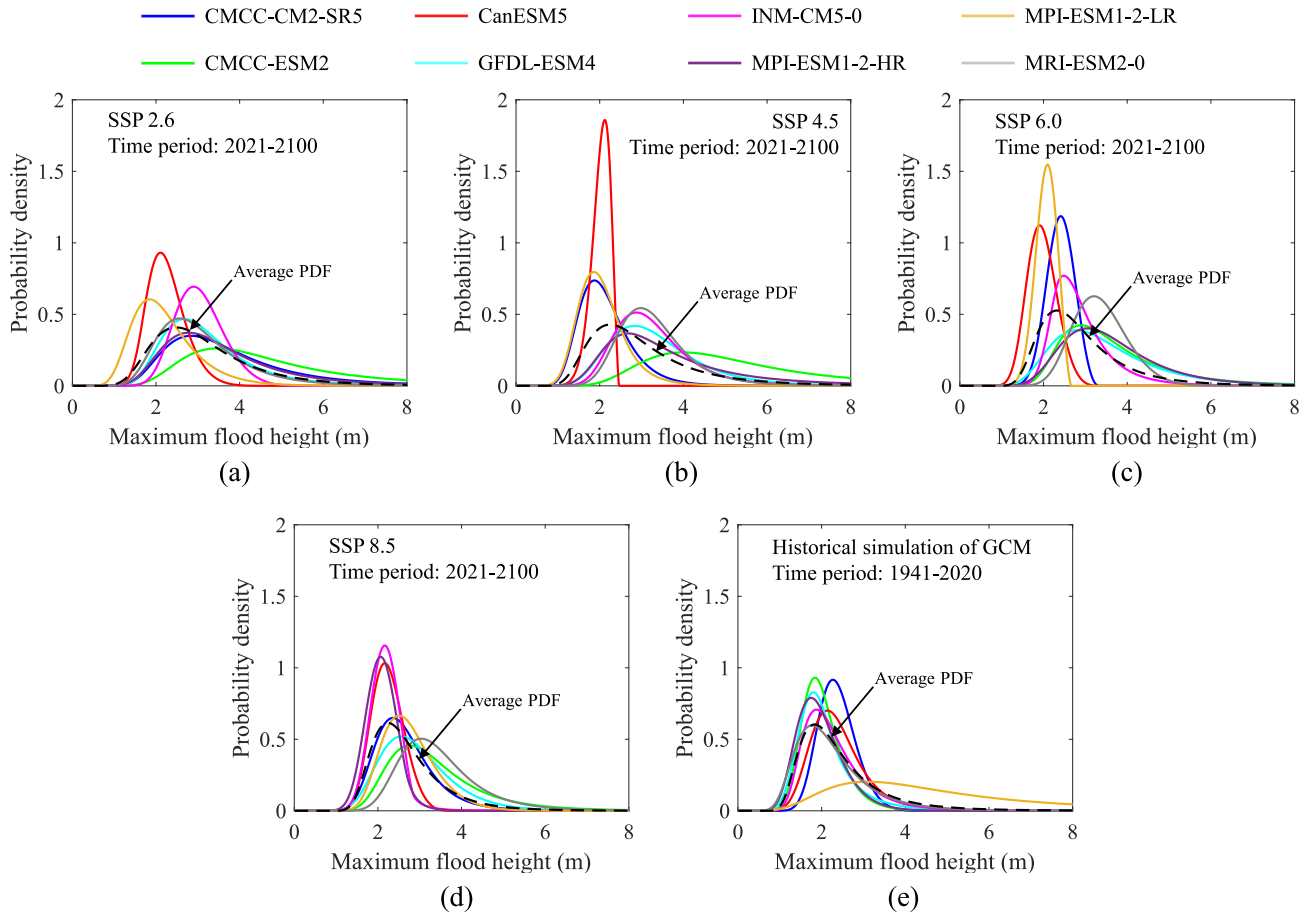


Fig. 15. PDF of maximum flood depth for each GCM under several climate change scenarios (a) SSP 2.6, (b) SSP 4.5, (c) SSP 6.0, (d) SSP 8.5, and (e) historical simulation of GCM from 1941 to 2020.

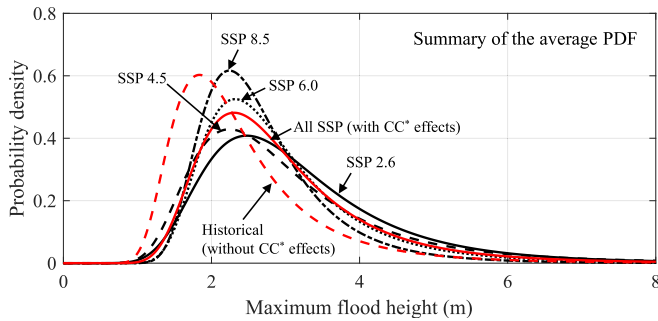


Fig. 16. Summary of the average PDF among GCMs for different SSPs in Fig. 17.

Traditionally, the assessment of maximum flood depth distribution relies on historical local weather records. To effectively evaluate the impact of climate change, these historical records from 1941 to 2020 (covering an 81-year period) can serve as a reference point for comparison, hereafter referred to as the probability of maximum flood depth without climate change effects. This alignment in time allows for a meaningful comparison between GCM simulations spanning from 2021 to 2100 and historical weather records. However, due to the limited availability of data in Fukuyama city, historical simulation data from GCMs are employed as a substitute for historical record data for illustrative purposes. Fig. 15(d) shows the PDF of the maximum flood depth using the historical simulation data of each GCM. It is evident that the average PDF of the maximum flood depth from the historical simulation

is notably lower than the future projections, with an expected value of 2.45 m. This distribution is used to estimate the partial factors γ_R^a and γ_S^a of the embankment against sliding. In addition, the partial factors considering the climate change effects (γ_R^b and γ_S^b) were evaluated using the average PDF of the maximum flood depth including all GCM and SSP (hereafter referred to as the PDF of the maximum flood depth with climate change effects), as shown in Fig. 15(e). Fig. 16 shows the distributions associated with the maximum flood depth $H_{exc,max}$ for both without and with climate change effects. The PDF for all SSPs in Fig. 16 was estimated assuming that the occurrence probability of each SSP is the same (i.e., 1/4).

5.4. Reliability assessment and partial factor design estimation of river embankment

The analyzed embankment is designed against overtopping and slope instability. This study focuses on developing a design methodology to achieve the target reliability for each limit state without considering the system reliability. Since the consequences of both limit states are relatively similar (i.e., failure of an embankment that causes flooding), the target reliability index β_t for both limit state functions is assumed to be the same. For the overtopping failure, the target reliability index β_t is used to determine the embankment design height as follows:

$$h_e = F_{H_{exc,max}}^{-1}(1 - \Phi_n(-\beta_t)) \quad (34)$$

After obtaining the embankment design height h_e , the reliability index against slope instability β_H and the associated MPP $\mathbf{X}^*$ are evaluated by solving the optimization problem in Eqs. (23) to (26). In this study, the set of random variables involved in the limit state function for

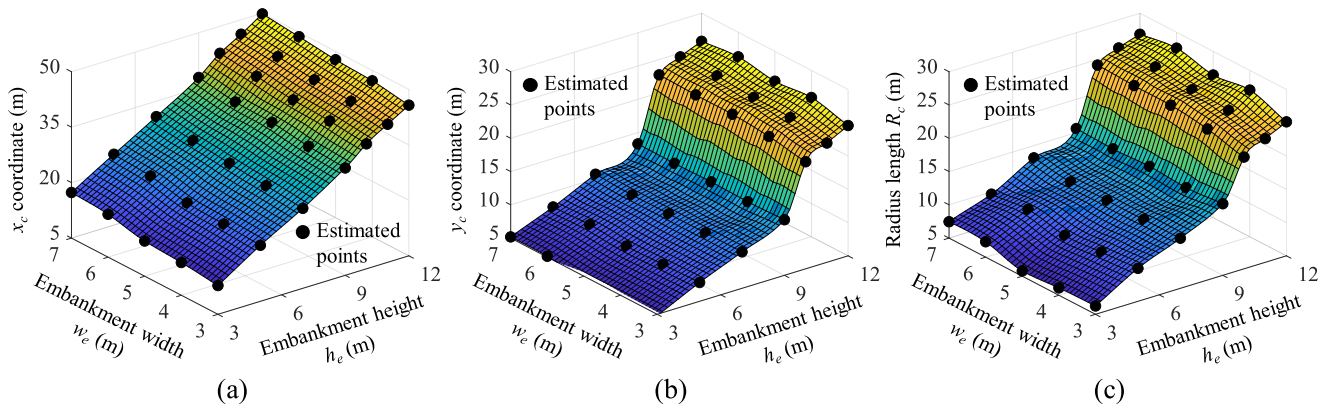


Fig. 17. Coordinates and radius length associated with the center of the critical failure surface relative to point A in Fig. 3: (a) x_c coordinate, (b) y_c coordinate, and (c) radius length R_c .

Table 1

Random variables associated with soil parameters and water height used in the case study.

Random variables	Distribution type	Mean (with CC ¹)	COV (with CC ¹)	Correlation coefficient	References
Dry density ρ_{dry}	Lognormal	16.5 (16.5)	0.10 (0.10)	−0.9	[109]
Effective cohesion c'	Lognormal	5.5 (5.5)	0.20 (0.20)		[110,111]
Effective friction angle ϕ'	Lognormal	32 (32)	0.20 (0.10)		[110,111]
Maximum flood depth $H_{exc,max}$	Eqs. (16) and (17)	2.48 (2.99)	0.40 (0.41)		

¹ CC = climate change.

slope instability is defined as $\mathbf{X} = \{c', \phi', \rho_{dry}, H_{exc,max}\}$, with $\rho_w = 10 \text{ kN/m}^3$ and $\rho_{sat} = 1.15\rho_{dry}$. In addition, the center coordinates of the critical failure surfaces and the corresponding radius are shown in Fig. 17 and were evaluated under an effective cohesion of 5 MPa, a friction angle of 32° , and a slope n_s of 2. The random variables associated with the soil parameters and the calculated maximum flood depth are summarized in Table 1. Finally, the design points X_D^* and the partial factors γ_X^a and γ_X^b were evaluated according to Fig. 6 and Eqs. (1) to (6).

Table 2 shows the reliability assessments and design for the random variables considering several target reliability indexes β_t . The increase in the target reliability index β_t necessitates a higher embankment design for the overtopping limit state, consequently increasing the volume of

sliding material and decreasing the reliability against slope instability. Furthermore, in response to climate change, the embankment height h_e should be raised since the distribution associated with the maximum flood depth $H_{exc,max}$ becomes more extreme. Notably, due to the negative correlation between the effective cohesion and the friction angle, the MPP for cohesion is greater than its mean value despite effective cohesion being strongly related to the resistance R in Eq. (20). Focusing on the partial factors to consider climate change effects γ_X^b , the soil parameters exhibits a lower value since these parameters are not directly influenced by climate change. In contrast, the maximum flood depth $H_{exc,max}$ stands out as the primary parameter significantly affected by climate change. The additional partial factor γ_X^b for the maximum flood

Table 2

Partial factor assessment for each design variable associated with the embankment.

Mean (with CC ¹)	Target reliability index	Design height of embankment		Reliability index from analysis		MPP	Design points		Partial factors	
μ_X ($\mu_{X,cc}$)	β_t	h_e	$h_{e,cc}$	β_H	$\beta_{H,cc}$	$\mathbf{X}^*$	X_D^*	$X_{D,cc}^*$	γ_X^a	γ_X^b
Friction angle ϕ (°)										
32	1.5	4.88	6.32	5.76	4.03	28.42	30.60	26.83	0.96	0.88
(32)	2	6.29	8.28	4.32	3.20	21.54	26.00	25.01	0.81	0.96
	2.5	8.34	11.11	3.56	3.05	23.15	24.43	25.27	0.76	1.03
Cohesion c (kPa)										
5.5	1.5	4.88	6.32	5.76	4.03	4.79	5.23	5.76	0.95	1.10
(5.5)	2	6.29	8.28	4.32	3.20	6.44	5.85	6.36	1.06	1.09
	2.5	8.34	11.11	3.56	3.05	6.75	6.47	6.41	1.18	0.99
Dry weight density ρ (kN/m3)										
16.5	1.5	4.88	6.32	5.76	4.03	10.59	14.41	15.49	0.87	1.08
(16.5)	2	6.29	8.28	4.32	3.20	14.11	15.26	15.61	0.92	1.02
	2.5	8.34	11.11	3.56	3.05	15.35	15.54	15.96	0.94	1.03
Maximum flood depth $H_{exc,max}$ (m)										
2.48	1.5	4.88	6.32	5.76	4.03	4.77	3.59	4.71	1.45	1.31
(2.99)	2	6.29	8.28	4.32	3.20	6.09	4.61	6.29	1.86	1.36
	2.5	8.34	11.11	3.56	3.05	7.52	6.38	8.96	2.57	1.40

¹ CC = climate change.

Table 3
Partial factors assessment for resistance R and load action S associated with slope instability limit state evaluated according to Table 2 and Eqs. (19) to (22).

Target reliability index	Design points of resistance R (kN/m)		Design points of load actions S (kN/m)		Mean value of resistance R (kN/m)		Mean value of load actions S (kN/m)		Partial factors of resistance R		Partial factors of load action S	
	R_D^*	$R_{D,cc}^*$	S_D^*	$S_{D,cc}^*$	R_k	$R_{k,cc}$	S_k	$S_{k,cc}$	γ_R^a	γ_R^b	γ_S^a	γ_S^b
2	294.6	409.7	196.3	318.5	393.6	549.7	218.0	326.7	0.75	1.39	0.90	1.62
2.5	394.2	551.3	310.4	486.6	558.3	796.0	321.2	487.3	0.71	1.40	0.97	1.57
3	544.7	644.8	489.5	613.0	817.5	962.9	489.2	593.8	0.67	1.18	1.00	1.25

depth ranges between 1.31 and 1.40, underscoring its vulnerability to climate-induced changes.

Table 3 shows the partial factors in resistance R and load actions S in the format of Eq. (4). The design points associated with the resistance R and load actions S were evaluated based on the design points X_D^* obtained from Table 3 according to Eqs. (20) and (21). The partial factor γ_R^a applied to the mean value of resistance R to ensure safety is below 1.0. However, the partial factor γ_S^a for load actions shows a value ranging between 0.9 and 1.0. This is primarily because the load action S is solely affected by the dry weight ρ_{dry} , where this random variable has a considerably lower COV value of 0.1. In addition, the dry weight ρ_{dry} appears in both resistance R and load actions S . Nevertheless, the additional partial factors γ_R^b and γ_S^b to accommodate the influence of climate change for both variables are substantially high. Verification of the results obtained in Table 3 is performed using MCS and is discussed in the supplementary files.

The results presented in this study indicate the importance of considering the effects of climate change on the design of embankments. In the context of flood protection for riverine embankments, extensive observational data collected from river stations in the future, particularly river height, can help to mitigate expectations and uncertainties in climate change, contributing to adjustments in design. This option may influence the choice of the initial target reliability. The proposed framework could serve as a reference for future code design to consider the impact of non-stationary climate change effects and should be adopted to promote a resilience community against riverine flooding disasters.

6. Conclusions

The conclusions resulting from this study are:

1. A novel framework for LRFD considering non-stationary climate change effects is established. Future precipitation and temperature are bias-corrected using the QDM method to create more accurate climate information.
2. A combination of the Pareto distribution and extreme value theory is developed to estimate the probability distribution of non-stationary maximum flood depth within a predefined time interval.
3. Hydrological and hydraulic analyses reveal that more severe SSP scenarios result in lower streamflow and flood depths due to higher temperatures associated with evapotranspiration. Nevertheless, compared to the historical simulations of GCMs, the PDF of maximum flood depth in the future is expected to be more extreme.
4. The reliability and partial factor estimation for river embankments are evaluated based on the SQP algorithm. Additional partial factors are introduced to transform the original design points into new design points that take into account the climate change effects.
5. The partial factors for resistance and load actions are substantially high, emphasizing the importance of considering climate change effects in the embankment design process.

The proposed framework can be extended to address various design problems related to non-stationary climate change effects, such as the degradation of infrastructure components [104–106], roof loads from

snow and rain [107], and long-term bridge scouring [108,112,113]. Another potential future direction of this research is the reliability-based design optimization application, where specific design parameters such as embankment steepness and height could be incorporated into the optimization problem with additional cost constraints. This study focused on evaluating embankment reliability under overtopping and slope instability. Future research should incorporate additional failure modes for more comprehensive reliability assessments and design, such as back erosion piping. In addition, while this study focuses on developing an LRFD methodology considering non-stationary climate change effects, future studies should consider the uncertainties associated with the hydrological and hydraulic modeling of the reliability-based design of embankments to achieve a more accurate result.

CRedit authorship contribution statement

Abdul Kadir Alhamid: Writing – original draft, Methodology, Investigation, Conceptualization. **Mitsuyoshi Akiyama:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Zhengying He:** Validation, Methodology, Investigation. **Putri Syahidah Firdaus:** Validation, Methodology, Investigation. **Dan M. Frangopol:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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